DiaTwin: AI-Powered Insulin Dosing

Profile for EXIST Open Consultation

1. Team & Skills

Contact Details

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Is Your Team Complete?

No. Actively recruiting 2 co-founders:
• **MD co-founder (Chief Medical Officer):** Medical Doctor, finished Aug 2025, completed practical year at Charité Berlin Interview completed, candidate very interested in the idea, currently discussing time commitment and role details , Final decision expected mid-February 2026
• **German pharmacist (COO):** 3 candidates in pipeline, target late February-March 2026
Missing skills: Clinical trial management, DiGA certification expertise, insurance negotiations

2. Motivation & Personal Drive

Why pursue this startup? Multiple family members have Type 1 Diabetes. I've watched them struggle with complex insulin calculations and fear dangerous blood sugar swings. As a pharmacist specializing in insulin therapy, I built DiaTwin during my master's thesis validated by Prof. Dr. Klaus Parhofer (LMU Munich Chief of Endocrinology) with 8.7/10 clinical utility rating and thesis grade 'Excellent.'

Why now?
(1) Germany's DiGA framework enables €1,800/year insurance reimbursement,
(2) 340,000 patients discontinued expensive sensors and lost precision dosing, untapped market,
(3) Prototype validated and ready for clinical pilot.

3. Startup Idea

Problem: Diabetes patients inject insulin 4-6 times daily but must guess doses manually. Errors cause dangerous high blood sugar (→ blindness, kidney failure) or low blood sugar (→ seizures, coma). Existing tools either cost €3,000-5,000/year (Pumps) or provide no dosing guidance (meters).

Target Group:

(1) 340,000 post-sensor users who discontinued expensive CGMs,
(2) 60,000 current sensor users needing dose calculation,
(3) 900,000 insulin pen users. Secondary: 5,000 endocrinologists needing efficient data review.

Solution: DiaTwin calculates exact insulin doses using smartphone + any glucose device (finger-prick or CGM). Works with ALL brands (Abbott, Dexcom, Medtronic, Accu-Chek). Tracks active insulin, adapts to exercise/meals, predicts blood sugar 4 hours ahead. Physician dashboard enables 5-minute patient review vs. 30-60 minutes.

Business Model: €150 one-time (CGM users) or €150 + €50/month (SMBG users). Physician dashboard: €50/month. DiGA certification = €1,800/year insurance reimbursement = €0 patient cost.

4. Current Status

Completed: Functional prototype (web + mobile), core algorithms tested with 50,000+ meal calculations, master's thesis validation (Grade 1.6 'Excellent' from LMU Munich), physician dashboard prototype.
Planned (EXIST Year): Clinical pilot study (20 patients, 3 months), CE marking + DiGA certification (18-24 months), GmbH formation, 50-100 beta testers.

5. Market & Competitors

Market Size: Germany: 1.3 million insulin users, 340,000 post-sensor patients (primary focus). Year 5 revenue potential: 2,000 users × €1,800 DiGA = €3.6M/year. Global: 50+ million insulin users in developing countries (freemium model).

Main Competitor: Accu-Chek SmartGuide

What it is: Android CGM reader (€399) with pre-installed mySugr app. Requires Accu-Chek SmartGuide CGM sensors (€80/month = €960/year). Total first year: €1,359.
Limitations: (1) Complete lock-in can't use Abbott Libre, Dexcom, or other CGMs,

(2) No physician dashboard, doctors spend 30-60 min reviewing PDFs,

Feature	SmartGuide	DiaTwin
Device compatibility	Accu-Chek only	**ANY device**
Physician dashboard	PDF export only	**5-min review**
Cost (Germany)	€1,359/year	**€0 (DiGA)**

Our Competitive Advantage: (1) Platform-agnostic works with existing hardware, no €399 reader purchase, (2) Post-sensor market focus 340,000 unserved patients, (3) Physician workflow integration saves 25 min/visit, (4) DiGA pathway = insurance coverage vs. €1,359 out-of-pocket.

6. Expectations & Key Questions

What I expect: (1) Team assessment is two-founders application viable with active recruitment pipeline? (2) Application strength evaluation, (3) Budget validation (€30K materials: €8K pilot study, €4K regulatory, €6.5K marketing), (4) Competitive positioning feedback.

Key questions: (1) Can co-founders be added during EXIST Year 1? (2) Is 20 patients × 3 months sufficient for DiGA approval? (3) what is the expected timeline of The grant?

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